KERALA POLYTECHNIC · REVISION 2026

Electric Vehicle Technology & Design Considerations

Complete student lesson for course code **5411**.

Revision 2026 Semester 5 Program Core 4 modules

Course snapshot

Scheme: 4 (L:3 T:1 P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 60

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Handbook method: Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

Course Dashboard

Course code

5411

Semester

5

Credits

4

Teaching scheme

4 (L:3 T:1 P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Electric and Hybrid Vehicle Fundamentals	15	CO1

No.	Module	Official hours	Relevant CO / level
2	EV Control Systems, Drive Trains and Braking	15	CO2
3	EV Design Requirements, Dynamics and Motors	15	CO3
4	EV Braking, Sensors, Cooling, Service and Safety	15	CO4

Prerequisites

- 2411 — Introduction to Electric Vehicles & Elementary Theory, semester 2; prerequisite topic: basic knowledge about storage batteries.
- 3411 — Electric Vehicle Batteries & Charging Systems, semester 3; prerequisite topic: basic knowledge of batteries and charging systems.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To understand general aspects of Electric and Hybrid Vehicles.
2. To understand the control system basics of EVs.
3. To understand basic design considerations of EVs.
4. To understand the basic concepts of braking systems and sensors.

Course Outcomes

1. CO1: Outline the Electric Vehicle and Hybrid Vehicle — 15 hours — Understanding.
2. CO2: Identify the various drive trains in EV & HEV — 15 hours — Understanding.
3. CO3: Outline the basic design considerations of EV — 15 hours — Remembering.

4. CO4: Outline the braking system for EVs — 13 hours — Understanding.

CO–PO mapping as supplied

CO1: PO1 = 2 (Moderately mapped).
CO2: PO1 = 2 (Moderately mapped).
CO3: PO1 = 2 (Moderately mapped).
CO4: PO1 = 2 (Moderately mapped).
Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	4	Official Contents block	8
Module 2	4	Official Contents block	9
Module 3	4	Official Contents block	4
Module 4	4	Official Contents block	5

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Need for hybrid and electric vehicles	4
module-1	M1.02	Components and working principles of electric and hybrid vehicles	4

Module	Topic code	Topic	Hours
module-1	M1.03	Configurations of Electric Vehicles	3
module-1	M1.04	Types of hybrid electric vehicle	4
module-2	M2.01	Control systems for EVs	4
module-2	M2.02	Split power devices for HEV	3
module-2	M2.03	Drive trains in EV and HEV	4
module-2	M2.04	Braking systems in EV	4
module-3	M3.01	Design requirements for electric vehicles	4
module-3	M3.02	Vehicle dynamics and performance	4
module-3	M3.03	Classification and requirements of EV motors	4
module-3	M3.04	Comparison of motors used in EVs	3
module-4	M4.01	Braking systems of EVs	3
module-4	M4.02	Regenerative braking and hybrid braking	4
module-4	M4.03	Sensors used in EVs	4
module-4	M4.04	Service and maintenance of EV components	4

Learning Roadmap

1. Electric and Hybrid Vehicle Fundamentals
CO1 · 15 hours

2. EV Control Systems, Drive Trains and Braking
CO2 · 15 hours

3. EV Design Requirements, Dynamics and Motors
CO3 · 15 hours

4. EV Braking, Sensors, Cooling, Service and Safety
CO4 · 15 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Electric and Hybrid Vehicle Fundamentals

This module establishes why electric and hybrid propulsion systems are used, how their energy and power paths are arranged, and how the principal vehicle configurations differ.

Hours: 15

CO1 · Bloom level: Understanding

Handbook study routine: Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 1

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.
Main components and working principles of electric and hybrid vehicles.
Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)
– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and
Without differential.
Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV (Extended range Electric Vehicle)

Point-by-point study guide

Official syllabus point 1: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.

Exact source point

Syllabus text: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.

How to understand and study it

This point is an official syllabus requirement: “Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Need for hybrid, electric vehicles, Historical background. Social, environmental importance of hybrid □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. using the official terminology retained above.
- Organise the important elements of Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.?
- How would you distinguish Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. from a closely related idea?
- Where would an engineer or technician encounter Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 2: Main components and working principles of electric and hybrid vehicles.

Exact source point

> **Syllabus text:** Main components and working principles of electric and hybrid vehicles.

How to understand and study it

This point is an official syllabus requirement: “Main components and working principles of electric and hybrid vehicles.”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Main components, working principles of electric, hybrid vehicles. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Main components and working principles of electric and hybrid vehicles. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Main components and working principles of electric and hybrid vehicles. using the official terminology retained above.
- Organise the important elements of Main components and working principles of electric and hybrid vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Main components and working principles of electric and hybrid vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Main components and working principles of electric and hybrid vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Main components and working principles of electric and hybrid vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Main components and working principles of electric and hybrid vehicles. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Main components and working principles of electric and hybrid vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Main components and working principles of electric and hybrid vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Main components and working principles of electric and hybrid vehicles.?
- How would you distinguish Main components and working principles of electric and hybrid vehicles. from a closely related idea?
- Where would an engineer or technician encounter Main components and working principles of electric and hybrid vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Main components and working principles of electric and hybrid vehicles. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Main components and working principles of electric and hybrid vehicles. □□□□ topic □□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 3: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery

Exact source point

Syllabus text: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery

How to understand and study it

This point is an official syllabus requirement: “Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery using the official terminology retained above.
- Organise the important elements of Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery into a logical sequence, classification, structure, or calculation.
- Connect Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery?
- How would you distinguish Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery from a closely related idea?
- Where would an engineer or technician encounter Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, and what must be checked before using it?
- What common mistake would make an answer or operation involving Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle

Exact source point

Syllabus text: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle

How to understand and study it

This point is an official syllabus requirement: “Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle using the official terminology retained above.
- Organise the important elements of Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle into a logical sequence, classification, structure, or calculation.
- Connect Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle?
- How would you distinguish Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle from a closely related idea?
- Where would an engineer or technician encounter Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle, and what must be checked before using it?
- What common mistake would make an answer or operation involving Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 5: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and

Exact source point

Syllabus text: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and

How to understand and study it

This point is an official syllabus requirement: “– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and using the official terminology retained above.
- Organise the important elements of – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and into a logical sequence, classification, structure, or calculation.
- Connect – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and?
- How would you distinguish – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and from a closely related idea?
- Where would an engineer or technician encounter – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and, and what must be checked before using it?
- What common mistake would make an answer or operation involving – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 6: Without differential.

Exact source point

> **Syllabus text:** Without differential.

How to understand and study it

This point is an official syllabus requirement: “Without differential.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Without differential. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Without differential. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Without differential. using the official terminology retained above.
- Organise the important elements of Without differential. into a logical sequence, classification, structure, or calculation.
- Connect Without differential. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Without differential. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Without differential.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Without differential. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Without differential., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Without differential., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Without differential.?
- How would you distinguish Without differential. from a closely related idea?
- Where would an engineer or technician encounter Without differential., and what must be checked before using it?
- What common mistake would make an answer or operation involving Without differential. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Without differential. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 7: Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV

Exact source point

> **Syllabus text:** Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV

How to understand and study it

This point is an official syllabus requirement: “Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ Types of HEV - series, parallel, series-parallel, Complex. Basic idea of EREV □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV using the official terminology retained above.
- Organise the important elements of Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV into a logical sequence, classification, structure, or calculation.
- Connect Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV?
- How would you distinguish Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV from a closely related idea?
- Where would an engineer or technician encounter Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV, and what must be checked before using it?
- What common mistake would make an answer or operation involving Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 8: (Extended range Electric Vehicle)

Exact source point

Syllabus text: (Extended range Electric Vehicle)

How to understand and study it

This point is an official syllabus requirement: “(Extended range Electric Vehicle)”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ (Extended range Electric Vehicle) □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

(Extended range Electric Vehicle) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope (Extended range Electric Vehicle) using the official terminology retained above.
- Organise the important elements of (Extended range Electric Vehicle) into a logical sequence, classification, structure, or calculation.
- Connect (Extended range Electric Vehicle) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on (Extended range Electric Vehicle) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading (Extended range Electric Vehicle). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of (Extended range Electric Vehicle) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on (Extended range Electric Vehicle), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is (Extended range Electric Vehicle), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining (Extended range Electric Vehicle)?
- How would you distinguish (Extended range Electric Vehicle) from a closely related idea?
- Where would an engineer or technician encounter (Extended range Electric Vehicle), and what must be checked before using it?
- What common mistake would make an answer or operation involving (Extended range Electric Vehicle) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

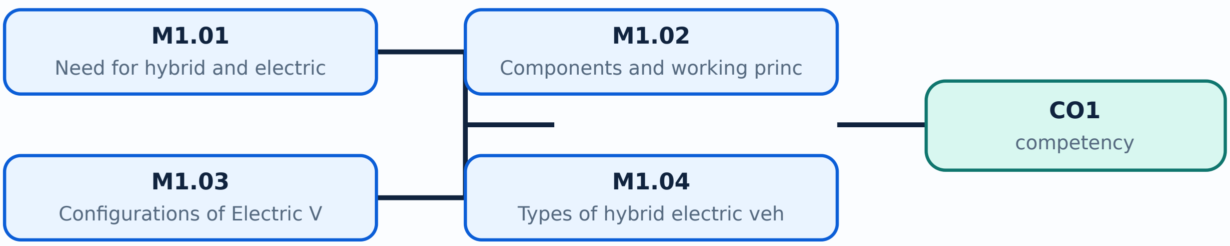
Malayalam support: (Extended range Electric Vehicle) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Learning goals

- Relate the need for EVs and HEVs to energy, environmental, and transport requirements.
- Identify the main components and trace the working principle of electric and hybrid vehicles.
- Differentiate BEV, FCEV, HEV, PHEV, EREV, and common HEV configurations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Need for hybrid and electric vehicles (4 hours)

Concept and explanation

Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and the possibility of using electricity from diverse sources. Historical development moved from early battery vehicles through modern power electronics, lithium-based batteries, charging infrastructure, and electronically controlled propulsion. Social importance includes quieter transport and reduced local pollution; environmental benefit depends on the electricity-generation mix, battery manufacture, and responsible end-of-life management.

Malayalam support: EV, HEV □□□□□□□□□□ □□□□□□□ energy efficiency, emission reduction, quieter operation, regenerative braking □□□□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□. Environmental benefit □□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□□; electricity source □□ battery life-cycle □□ □□□□□□□□□□□.

Study points

- State at least three technical or environmental reasons for adopting EVs or HEVs.
- Distinguish local tail-pipe emissions from total life-cycle environmental impact.
- Mention the role of batteries, motors, power electronics, and charging infrastructure.

Engineering / workplace application: Electric mobility is used in passenger cars, buses, delivery vehicles, forklifts, and two-wheelers where efficient stop-start operation is valuable.

Illustrative example: If a vehicle recovers 20 kJ during braking instead of dissipating it as heat, that recovered energy can be returned to the battery through the power electronics and motor-generator path.

Common mistake: Claiming that every EV has zero total environmental impact ignores electricity generation and battery manufacturing.

Handbook treatment

M1.01 — Need for hybrid and electric vehicles (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 — Need for hybrid and electric vehicles (4 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — Need for hybrid and electric vehicles (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — Need for hybrid and electric vehicles (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.01 — Need for hybrid and electric vehicles (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 — Need for hybrid and electric vehicles (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 — Need for hybrid and electric vehicles (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 — Need for hybrid and electric vehicles (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 — Need for hybrid and electric vehicles (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — Need for hybrid and electric vehicles (4 hours)?
- How would you distinguish M1.01 — Need for hybrid and electric vehicles (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — Need for hybrid and electric vehicles (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — Need for hybrid and electric vehicles (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 — Need for hybrid and electric vehicles (4 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.02 — Components and working principles of electric and hybrid vehicles (4 hours)

Concept and explanation

A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loads, and vehicle control unit. An HEV adds an engine, fuel system, generator or motor-generator, and a supervisory energy-management controller. During propulsion, stored electrical energy passes through the inverter to the motor; during regenerative deceleration, the power flow reverses and the motor operates as a generator.

Malayalam support: Battery, inverter, traction motor, controller, DC-DC converter, charger, differential □□□□□□□□□□ EV-□□□□ □□□□□□ parts. HEV-□ engine □□□□ □□□□□□□□□. Propulsion □□□□□□□ energy battery□□□ □□□□□□ motor□□□□□□□ regenerative braking □□□□□□□ motor□ □□□□□□ battery□□□□□□□□□ □□□□□□□□.

Study points

- Draw the energy path from battery to wheel and the reverse path during regeneration.
- Explain the different roles of the inverter, DC-DC converter, charger, and vehicle control unit.
- Compare the additional components required by an HEV.

Engineering / workplace application: The same energy-flow reasoning is used when diagnosing abnormal range, inverter temperature, motor torque, or battery state-of-charge behaviour.

Illustrative example: For a BEV, battery DC is converted by the inverter into controlled three-phase AC for a traction motor; the reduction gear converts motor speed and torque to wheel speed and tractive torque.

Common mistake: The onboard charger is not the same as the traction inverter; one primarily converts charging power while the other controls motor power during driving.

Handbook treatment

M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 — Components and working principles of electric and hybrid vehicles (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.02 — Components and working principles of electric and hybrid vehicles (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 — Components and working principles of electric and hybrid vehicles (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Components and working principles of electric and hybrid vehicles (4 hours)?
- How would you distinguish M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Components and working principles of electric and hybrid vehicles (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.02 — Components and working principles of electric and hybrid vehicles (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.03 — Configurations of Electric Vehicles (3 hours)

Concept and explanation

The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An FCEV uses a fuel cell to generate electricity, usually with a battery or buffer storage. An HEV combines an engine and an electrical path, while a PHEV adds a larger rechargeable battery and external charging capability. EVs may also be classified by whether torque is transmitted through a differential or through separate in-wheel or near-wheel drives.

Malayalam support: BEV, FCEV, HEV, PHEV □□□□□ □□□□□□ energy source, charging method, engine presence, drive arrangement □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□. Differential □□□□□□□□ in-wheel drive □ □□□ wheel-□□□ □□□□□□□□□ motor control □□□□□□□□□□□□.

Study points

- Prepare a comparison using energy source, external charging, engine presence, and typical operating mode.
- Explain the meaning of with-differential and without-differential classifications.
- Use the official abbreviations correctly in answers.

Engineering / workplace application: Vehicle configuration affects packaging, unsprung mass, control complexity, service procedures, and energy efficiency.

Illustrative example: A BEV has no combustion engine for normal propulsion, whereas a PHEV can operate in electric mode and later use its engine when battery energy or power demand requires it.

Common mistake: PHEV is not simply another name for BEV; a PHEV retains an engine and can normally be externally charged.

Handbook treatment

M1.03 — Configurations of Electric Vehicles (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.03 — Configurations of Electric Vehicles (3 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Configurations of Electric Vehicles (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Configurations of Electric Vehicles (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.03 — Configurations of Electric Vehicles (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 — Configurations of Electric Vehicles (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.03 — Configurations of Electric Vehicles (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.03 — Configurations of Electric Vehicles (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 — Configurations of Electric Vehicles (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Configurations of Electric Vehicles (3 hours)?
- How would you distinguish M1.03 — Configurations of Electric Vehicles (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Configurations of Electric Vehicles (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Configurations of Electric Vehicles (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.03 — Configurations of Electric Vehicles (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.04 — Types of hybrid electric vehicle (4 hours)

Concept and explanation

In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split device. A complex hybrid can include multiple motors, clutches, or flexible power paths. An Extended Range Electric Vehicle (EREV) primarily uses the electric drive for wheel propulsion while an engine-generator extends the available range when required.

Malayalam support: Series HEV □ engine generator □□□□□□□□, motor wheels □□□□□□□□. Parallel HEV □ engine□□□ motor□□□ wheel torque □□□□□. Series-parallel power split device □□□□□□□□□□ □□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□. EREV-□ engine □□□□□□□□□□ range extender □□□ □□□□□□□□□□□□□□□.

Study points

- Trace mechanical and electrical power paths for series, parallel, and series-parallel HEVs.
- State one advantage and one limitation of each major hybrid arrangement.
- Distinguish an EREV range-extender role from direct mechanical engine drive.

Engineering / workplace application: Powertrain architecture is selected according to required acceleration, cruising efficiency, packaging, cost, and control strategy.

Illustrative example: In a series HEV, the engine can run near an efficient operating point to generate electricity while the traction motor responds directly to wheel torque demand.

Common mistake: Calling a series HEV a direct mechanical engine-to-wheel system reverses its defining power-flow arrangement.

Handbook treatment

M1.04 — Types of hybrid electric vehicle (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.04 — Types of hybrid electric vehicle (4 hours) using the official terminology retained above.
- Organise the important elements of M1.04 — Types of hybrid electric vehicle (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.04 — Types of hybrid electric vehicle (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.04 — Types of hybrid electric vehicle (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 — Types of hybrid electric vehicle (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.04 — Types of hybrid electric vehicle (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.04 — Types of hybrid electric vehicle (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 — Types of hybrid electric vehicle (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 — Types of hybrid electric vehicle (4 hours)?
- How would you distinguish M1.04 — Types of hybrid electric vehicle (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.04 — Types of hybrid electric vehicle (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 — Types of hybrid electric vehicle (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.04 — Types of hybrid electric vehicle (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: EV and HEV classification becomes clear when the student follows the energy source, conversion path, and wheel-torque path.

OFFICIAL MODULE 2

EV Control Systems, Drive Trains and Braking

This module connects vehicle requirements to drive-system sizing, control, drive-train architecture, and braking. The central idea is coordinated power flow between energy storage, electronics, machines, wheels, and auxiliary systems.

Hours: 15

CO2 · Bloom level: Understanding

Handbook study routine: Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 2

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Control systems for EV - Sizing the drive system - Sizing the propulsion motor, sizing the power electronics - Selecting the energy storage technology - Communications -supporting subsystems. Power split devices for Hybrid Vehicles.
Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In
Wheel Drives.
Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

Point-by-point study guide

Official syllabus point 1: Control systems for EV

Exact source point

Syllabus text: Control systems for EV

How to understand and study it

This point is an official syllabus requirement: “Control systems for EV”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Control systems for EV □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Control systems for EV should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope Control systems for EV using the official terminology retained above.
- Organise the important elements of Control systems for EV into a logical sequence, classification, structure, or calculation.
- Connect Control systems for EV with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Control systems for EV with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Control systems for EV. It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of Control systems for EV depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.
Required: a suitable measurement chain or an explanation of how the instrument operates.
Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on Control systems for EV, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Control systems for EV, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Control systems for EV?
- How would you distinguish Control systems for EV from a closely related idea?
- Where would an engineer or technician encounter Control systems for EV, and what must be checked before using it?
- What common mistake would make an answer or operation involving Control systems for EV incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: Control systems for EV [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 2: Sizing the drive system

Exact source point

Syllabus text: Sizing the drive system

How to understand and study it

This point is an official syllabus requirement: “Sizing the drive system”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Sizing the drive system □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Sizing the drive system is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Sizing the drive system using the official terminology retained above.
- Organise the important elements of Sizing the drive system into a logical sequence, classification, structure, or calculation.
- Connect Sizing the drive system with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Sizing the drive system with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Sizing the drive system. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Sizing the drive system is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Sizing the drive system, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Sizing the drive system, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Sizing the drive system?
- How would you distinguish Sizing the drive system from a closely related idea?
- Where would an engineer or technician encounter Sizing the drive system, and what must be checked before using it?
- What common mistake would make an answer or operation involving Sizing the drive system incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Sizing the drive system [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: Sizing the propulsion motor, sizing the power electronics

Exact source point

Syllabus text: Sizing the propulsion motor, sizing the power electronics

How to understand and study it

This point is an official syllabus requirement: “Sizing the propulsion motor, sizing the power electronics”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Sizing the propulsion motor, sizing the power electronics □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Sizing the propulsion motor, sizing the power electronics must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Sizing the propulsion motor, sizing the power electronics using the official terminology retained above.
- Organise the important elements of Sizing the propulsion motor, sizing the power electronics into a logical sequence, classification, structure, or calculation.
- Connect Sizing the propulsion motor, sizing the power electronics with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Sizing the propulsion motor, sizing the power electronics with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Sizing the propulsion motor, sizing the power electronics. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Sizing the propulsion motor, sizing the power electronics is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Sizing the propulsion motor, sizing the power electronics, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Sizing the propulsion motor, sizing the power electronics, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Sizing the propulsion motor, sizing the power electronics?
- How would you distinguish Sizing the propulsion motor, sizing the power electronics from a closely related idea?
- Where would an engineer or technician encounter Sizing the propulsion motor, sizing the power electronics, and what must be checked before using it?
- What common mistake would make an answer or operation involving Sizing the propulsion motor, sizing the power electronics incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Sizing the propulsion motor, sizing the power electronics [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: Selecting the energy storage technology

Exact source point

Syllabus text: Selecting the energy storage technology

How to understand and study it

This point is an official syllabus requirement: “Selecting the energy storage technology”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Selecting the energy storage technology □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Selecting the energy storage technology must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Selecting the energy storage technology using the official terminology retained above.
- Organise the important elements of Selecting the energy storage technology into a logical sequence, classification, structure, or calculation.
- Connect Selecting the energy storage technology with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Selecting the energy storage technology with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Selecting the energy storage technology. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Selecting the energy storage technology is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Selecting the energy storage technology, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Selecting the energy storage technology, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Selecting the energy storage technology?
- How would you distinguish Selecting the energy storage technology from a closely related idea?
- Where would an engineer or technician encounter Selecting the energy storage technology, and what must be checked before using it?
- What common mistake would make an answer or operation involving Selecting the energy storage technology incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Selecting the energy storage technology [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 5: Communications -supporting subsystems. Power split devices for Hybrid Vehicles.

Exact source point

Syllabus text: Communications -supporting subsystems. Power split devices for Hybrid Vehicles.

How to understand and study it

This point is an official syllabus requirement: “Communications -supporting subsystems. Power split devices for Hybrid Vehicles.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Communications -supporting subsystems. Power split devices for Hybrid Vehicles. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Communications -supporting subsystems. Power split devices for Hybrid Vehicles. must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Communications -supporting subsystems. Power split devices for Hybrid Vehicles. using the official terminology retained above.
- Organise the important elements of Communications -supporting subsystems. Power split devices for Hybrid Vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Communications -supporting subsystems. Power split devices for Hybrid Vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Communications -supporting subsystems. Power split devices for Hybrid Vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Communications -supporting subsystems. Power split devices for Hybrid Vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Communications -supporting subsystems. Power split devices for Hybrid Vehicles. is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Communications -supporting subsystems. Power split devices for Hybrid Vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Communications -supporting subsystems. Power split devices for Hybrid Vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Communications -supporting subsystems. Power split devices for Hybrid Vehicles.?
- How would you distinguish Communications -supporting subsystems. Power split devices for Hybrid Vehicles. from a closely related idea?
- Where would an engineer or technician encounter Communications -supporting subsystems. Power split devices for Hybrid Vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Communications -supporting subsystems. Power split devices for Hybrid Vehicles. incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Communications -supporting subsystems. Power split devices for Hybrid Vehicles. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 6: Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration

Exact source point

> **Syllabus text:** Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration

How to understand and study it

This point is an official syllabus requirement: “Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Drive train in EV, Basic Architecture of Electric Drive Trains, General configuration of an electric vehicle, Alternatives Based on Drive train Configuration □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration using the official terminology retained above.
- Organise the important elements of Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration into a logical sequence, classification, structure, or calculation.
- Connect Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on

Drive train Configuration is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration?
- How would you distinguish Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration from a closely related idea?
- Where would an engineer or technician encounter Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, and what must be checked before using it?

- What common mistake would make an answer or operation involving Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

> **Malayalam support:** Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 7: Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In

Exact source point

> **Syllabus text:** Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In

How to understand and study it

This point is an official syllabus requirement: “Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ Alternatives Based on Power Source Configuration, Single, Multi-motor Drives - In □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In using the official terminology retained above.
- Organise the important elements of Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In into a logical sequence, classification, structure, or calculation.
- Connect Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In?
- How would you distinguish Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In from a closely related idea?
- Where would an engineer or technician encounter Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, and what must be checked before using it?
- What common mistake would make an answer or operation involving Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 8: Wheel Drives.

Exact source point

Syllabus text: Wheel Drives.

How to understand and study it

This point is an official syllabus requirement: “Wheel Drives.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Wheel Drives. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Wheel Drives. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Wheel Drives. using the official terminology retained above.
- Organise the important elements of Wheel Drives. into a logical sequence, classification, structure, or calculation.
- Connect Wheel Drives. with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Wheel Drives. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Wheel Drives.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Wheel Drives. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Wheel Drives., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Wheel Drives., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Wheel Drives.?
- How would you distinguish Wheel Drives. from a closely related idea?
- Where would an engineer or technician encounter Wheel Drives., and what must be checked before using it?
- What common mistake would make an answer or operation involving Wheel Drives. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Wheel Drives. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 9: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

Exact source point

Syllabus text: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

How to understand and study it

This point is an official syllabus requirement: “Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Braking system in electric, Hybrid vehicles, Regenerative braking, Hybrid braking. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. using the official terminology retained above.
- Organise the important elements of Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. into a logical sequence, classification, structure, or calculation.
- Connect Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.?
- How would you distinguish Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. from a closely related idea?
- Where would an engineer or technician encounter Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., and what must be checked before using it?
- What common mistake would make an answer or operation involving Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

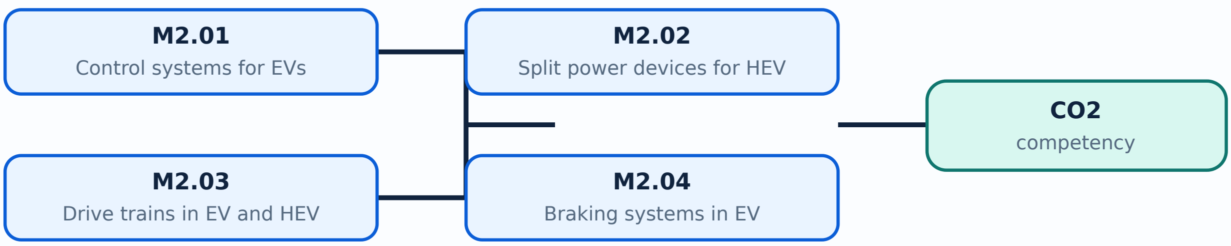
Malayalam support: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Learning goals

- List the principal functions of an EV control system and supporting subsystem.
- Compare power-split devices and single-, multi-, and in-wheel motor drive trains.
- Explain friction, regenerative, and hybrid braking at a system level.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Control systems for EVs (4 hours)

Concept and explanation

An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-electronics rating, energy-storage selection, communications network, and supporting subsystems such as cooling, steering, lighting, and braking. The controller must enforce voltage, current, temperature, and isolation limits while meeting the requested vehicle torque.

Malayalam support: EV control system accelerator input, battery state-of-charge, torque, temperature, charging, regenerative braking □□□□□□ □□□□□□□□□□□□□□□□□. Sizing □□□□□□□□□□□ motor, power electronics, energy storage, communication, cooling □□□□□□ □□□□□□□□□ □□□□□□□□□□□□.

Study points

- Separate energy capacity requirements from peak power requirements.
- Identify feedback signals and protection limits used by a vehicle controller.
- Explain why motor, inverter, battery, and thermal sizing cannot be designed independently.

Engineering / workplace application: Battery-management and vehicle-control systems are central to safe charging, predictable range, and coordinated torque in production EVs.

Illustrative example: If the driver requests acceleration, the controller checks battery and temperature limits, commands inverter current, and reduces torque when a protection threshold is reached.

Common mistake: Selecting a large battery alone does not guarantee high acceleration; peak current, inverter power, motor torque, and thermal limits also matter.

Handbook treatment

M2.01 — Control systems for EVs (4 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope M2.01 — Control systems for EVs (4 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Control systems for EVs (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Control systems for EVs (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on M2.01 — Control systems for EVs (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 — Control systems for EVs (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of M2.01 — Control systems for EVs (4 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.
Required: a suitable measurement chain or an explanation of how the instrument operates.
Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on M2.01 — Control systems for EVs (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 — Control systems for EVs (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Control systems for EVs (4 hours)?
- How would you distinguish M2.01 — Control systems for EVs (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Control systems for EVs (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Control systems for EVs (4 hours) incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: M2.01 — Control systems for EVs (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.02 — Split power devices for HEV (3 hours)

Concept and explanation

A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator and motor torque so the engine can operate efficiently while the vehicle meets its speed and acceleration demand.

Malayalam support: Power-split device engine, generator, motor, wheels □□□□□□□□□□□□□□□□□□ power flow □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□ □□□□□□□□. Planetary gear set □ speed-torque relationship □□□□□□ engine efficient region □ □□□□□□□□□□□□□□□□□□□.

Study points

- Name the purpose of a power-split device.
- Explain that speed and torque are coupled through the gear arrangement.
- Relate generator control to engine operating-point management.

Engineering / workplace application: Power-split systems are used in hybrid transmissions to blend mechanical and electrical power during launch, cruising, and regeneration.

Illustrative example: During low-speed launch, the motor can provide wheel torque while the engine remains off or operates through the generator path.

Common mistake: A power-split device is not itself a battery; it is a mechanical/electromechanical arrangement that controls paths of power.

Handbook treatment

M2.02 — Split power devices for HEV (3 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M2.02 — Split power devices for HEV (3 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Split power devices for HEV (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Split power devices for HEV (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on M2.02 — Split power devices for HEV (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 — Split power devices for HEV (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M2.02 — Split power devices for HEV (3 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M2.02 — Split power devices for HEV (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 — Split power devices for HEV (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Split power devices for HEV (3 hours)?
- How would you distinguish M2.02 — Split power devices for HEV (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Split power devices for HEV (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Split power devices for HEV (3 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M2.02 — Split power devices for HEV (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.03 — Drive trains in EV and HEV (4 hours)

Concept and explanation

An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery, fuel-cell, hybrid, or multiple-source power. Single-motor drives are simpler; multi-motor drives can distribute torque between axles or wheels. In-wheel drives reduce mechanical transmission parts but increase wheel mass and expose the motor to harsh conditions.

Malayalam support: Drive train □ source, converter, motor, reduction gear, differential □□□□□□□□□□ wheel motors, wheels □□□□□□ □□□□□□□□□□□□□. Single motor □□□□□□□□; multi-motor torque distribution □□□□□□□□. In-wheel motor transmission parts □□□□□□□□□□, □□□□□ unsprung mass □□□□□.

Study points

- Compare single-motor, multi-motor, and in-wheel drive arrangements.
- Explain how the power source changes the overall drive-train architecture.
- Mention packaging, control, efficiency, and service trade-offs.

Engineering / workplace application: Drive-train choice influences traction control, vehicle stability, packaging, and maintenance in buses, cars, and industrial vehicles.

Illustrative example: An all-wheel-drive EV can command different axle torques to improve traction on a low-friction surface.

Common mistake: More motors do not automatically mean better efficiency; converter losses, mass, control complexity, and operating point must be considered.

Handbook treatment

M2.03 — Drive trains in EV and HEV (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.03 — Drive trains in EV and HEV (4 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — Drive trains in EV and HEV (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — Drive trains in EV and HEV (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on M2.03 — Drive trains in EV and HEV (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 — Drive trains in EV and HEV (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.03 — Drive trains in EV and HEV (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.03 — Drive trains in EV and HEV (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 — Drive trains in EV and HEV (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — Drive trains in EV and HEV (4 hours)?
- How would you distinguish M2.03 — Drive trains in EV and HEV (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — Drive trains in EV and HEV (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — Drive trains in EV and HEV (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.03 — Drive trains in EV and HEV (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.04 — Braking systems in EV (4 hours)

Concept and explanation

EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provide the remaining deceleration and hold the vehicle at rest. Hybrid braking coordinates pedal feel, motor torque, hydraulic pressure, ABS, and stability control. The braking system must remain effective when the battery is full, cold, overheated, or unable to accept regeneration.

Malayalam support: Regenerative braking energy battery□□□□□□□□ □□□□□□□, □□□□□ battery full/cold/temperature limit □□□□□□□□□ friction brake □□□□□□□□□. Hybrid braking motor torque, hydraulic pressure, ABS, stability control □□□□□□ coordinate □□□□□□□□□□.

Study points

- Explain why regenerative braking cannot provide all braking in every condition.
- Differentiate regenerative, friction, and blended or hybrid braking.
- Relate braking control to ABS and battery acceptance limits.

Engineering / workplace application: Blended braking improves efficiency while maintaining reliable stopping distance in road vehicles.

Illustrative example: During a gentle stop with available battery capacity, the controller may request generator torque first and add hydraulic braking if the requested deceleration is not achieved.

Common mistake: Assuming regeneration is available during every braking event can lead to unsafe design and incorrect energy estimates.

Handbook treatment

M2.04 — Braking systems in EV (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.04 — Braking systems in EV (4 hours) using the official terminology retained above.
- Organise the important elements of M2.04 — Braking systems in EV (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.04 — Braking systems in EV (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Control Systems, Drive Trains and Braking.
- Write an examination answer on M2.04 — Braking systems in EV (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 — Braking systems in EV (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.04 — Braking systems in EV (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.04 — Braking systems in EV (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 — Braking systems in EV (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 — Braking systems in EV (4 hours)?
- How would you distinguish M2.04 — Braking systems in EV (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.04 — Braking systems in EV (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 — Braking systems in EV (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.04 — Braking systems in EV (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: Control, drive-train, and brake decisions are coupled: the vehicle must satisfy torque and safety requirements while managing energy flow.

OFFICIAL MODULE 3

EV Design Requirements, Dynamics and Motors

This module gives the design language for vehicle range, speed, acceleration, resistive forces, tractive effort, transmission, energy consumption, and motor selection.

Hours: 15
CO3 · Bloom level: Remembering / Understanding

Handbook study routine: Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 3

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

:
Design requirement for electric vehicles- Range, maximum velocity, acceleration, power
Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive
Effort and Transmission. Requirement, Transmission efficiency, energy consumption.
Classification & Requirements of EV motors - Brushed & Brushless DC motors ,
Electric motor comparison

Point-by-point study guide

Official syllabus point 1: Design requirement for electric vehicles- Range, maximum velocity, acceleration, power

Exact source point

> **Syllabus text:** Design requirement for electric vehicles- Range, maximum velocity, acceleration, power

How to understand and study it

This point is an official syllabus requirement: “Design requirement for electric vehicles- Range, maximum velocity, acceleration, power”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Design requirement for electric vehicles- Range, maximum velocity, acceleration □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Design requirement for electric vehicles- Range, maximum velocity, acceleration, power must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Design requirement for electric vehicles- Range, maximum velocity, acceleration, power using the official terminology retained above.
- Organise the important elements of Design requirement for electric vehicles- Range, maximum velocity, acceleration, power into a logical sequence, classification, structure, or calculation.
- Connect Design requirement for electric vehicles- Range, maximum velocity, acceleration, power with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on Design requirement for electric vehicles- Range, maximum velocity, acceleration, power with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Design requirement for electric vehicles- Range, maximum velocity, acceleration, power. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Design requirement for electric vehicles- Range, maximum velocity, acceleration, power is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Design requirement for electric vehicles- Range, maximum velocity, acceleration, power, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Design requirement for electric vehicles- Range, maximum velocity, acceleration, power, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Design requirement for electric vehicles- Range, maximum velocity, acceleration, power?
- How would you distinguish Design requirement for electric vehicles- Range, maximum velocity, acceleration, power from a closely related idea?
- Where would an engineer or technician encounter Design requirement for electric vehicles- Range, maximum velocity, acceleration, power, and what must be checked before using it?
- What common mistake would make an answer or operation involving Design requirement for electric vehicles- Range, maximum velocity, acceleration, power incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Design requirement for electric vehicles- Range, maximum velocity, acceleration, power [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 2: Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive

Exact source point

> **Syllabus text:** Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive

How to understand and study it

This point is an official syllabus requirement: “Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□□ Requirement, mass of the vehicle. Various Resistances- rolling resistance, aerodynamic drag- Performance of electric vehicles acceleration, moving up □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive using the official terminology retained above.
- Organise the important elements of Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive into a logical sequence, classification, structure, or calculation.
- Connect Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive?
- How would you distinguish Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive from a closely related idea?
- Where would an engineer or technician encounter Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag-

Performance of electric vehicles acceleration, moving up and down a hill. Tractive, and what must be checked before using it?

- What common mistake would make an answer or operation involving Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

> **Malayalam support:** Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□□.

Official syllabus point 3: Effort and Transmission. Requirement, Transmission efficiency, energy consumption.

Exact source point

Syllabus text: Effort and Transmission. Requirement, Transmission efficiency, energy consumption.

How to understand and study it

This point is an official syllabus requirement: “Effort and Transmission. Requirement, Transmission efficiency, energy consumption.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Effort, Transmission. Requirement, Transmission efficiency, energy consumption. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Effort and Transmission. Requirement, Transmission efficiency, energy consumption. must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Effort and Transmission. Requirement, Transmission efficiency, energy consumption. using the official terminology retained above.
- Organise the important elements of Effort and Transmission. Requirement, Transmission efficiency, energy consumption. into a logical sequence, classification, structure, or calculation.
- Connect Effort and Transmission. Requirement, Transmission efficiency, energy consumption. with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on Effort and Transmission. Requirement, Transmission efficiency, energy consumption. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Effort and Transmission. Requirement, Transmission efficiency, energy consumption.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Effort and Transmission. Requirement, Transmission efficiency, energy consumption. is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Effort and Transmission. Requirement, Transmission efficiency, energy consumption., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Effort and Transmission. Requirement, Transmission efficiency, energy consumption., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Effort and Transmission. Requirement, Transmission efficiency, energy consumption.?
- How would you distinguish Effort and Transmission. Requirement, Transmission efficiency, energy consumption. from a closely related idea?
- Where would an engineer or technician encounter Effort and Transmission. Requirement, Transmission efficiency, energy consumption., and what must be checked before using it?
- What common mistake would make an answer or operation involving Effort and Transmission. Requirement, Transmission efficiency, energy consumption. incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Effort and Transmission. Requirement, Transmission efficiency, energy consumption. □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□□.

Official syllabus point 4: Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison

Exact source point

Syllabus text: Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison

How to understand and study it

This point is an official syllabus requirement: “Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Classification & Requirements of EV motors - Brushed & Brushless DC motors, Electric motor comparison □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison using the official terminology retained above.
- Organise the important elements of Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison into a logical sequence, classification, structure, or calculation.
- Connect Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison?
- How would you distinguish Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison from a closely related idea?
- Where would an engineer or technician encounter Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison, and what must be checked before using it?
- What common mistake would make an answer or operation involving Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.

- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

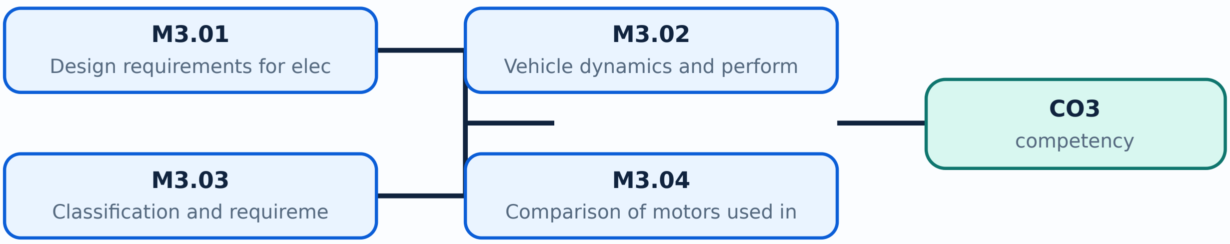
Malayalam support: Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Learning goals

- List the main performance requirements used in EV design.
- Calculate or identify rolling, aerodynamic, grade, and acceleration components of tractive effort.
- Classify EV motors and compare brushed and brushless DC motors with other practical choices.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Design requirements for electric vehicles (4 hours)

Concept and explanation

Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy storage can increase mass, while higher acceleration demands higher peak motor and inverter power. A design statement should define the duty cycle, road conditions, payload, ambient temperature, and allowable state-of-charge window before component ratings are selected.

Malayalam support: EV design □ range, maximum velocity, acceleration, gradeability, power, mass, charging time, safety, reliability □□□□□□ □□□□□□□□□□. Battery capacity □□□□□□□□□□□ mass □□□□□□; □□□□□□ requirements □□□□□□□□□□ trade-off □□□□□□□□.

Study points

- Differentiate range, peak power, continuous power, and gradeability requirements.
- State the operating conditions that must be fixed before rating components.
- Explain the trade-off between battery mass and range.

Engineering / workplace application: Fleet duty-cycle data is used to size batteries and motors for buses, delivery vans, and industrial vehicles.

Illustrative example: A city bus with frequent stops may prioritise high regenerative capability and peak torque, while a highway vehicle may prioritise continuous power and aerodynamic efficiency.

Common mistake: Using only maximum speed to size an EV ignores launch torque, hill climbing, payload, thermal duty, and range.

Handbook treatment

M3.01 — Design requirements for electric vehicles (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.01 — Design requirements for electric vehicles (4 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Design requirements for electric vehicles (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Design requirements for electric vehicles (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on M3.01 — Design requirements for electric vehicles (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 — Design requirements for electric vehicles (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.01 — Design requirements for electric vehicles (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.01 — Design requirements for electric vehicles (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 — Design requirements for electric vehicles (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Design requirements for electric vehicles (4 hours)?
- How would you distinguish M3.01 — Design requirements for electric vehicles (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Design requirements for electric vehicles (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Design requirements for electric vehicles (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.01 — Design requirements for electric vehicles (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

M3.02 — Vehicle dynamics and performance (4 hours)

Concept and explanation

The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is F_trac = F_roll + F_aero + F_grade + m a. Rolling resistance may be approximated by C_rr m g; aerodynamic drag by 0.5 rho C_d A v^2; grade force by m g sin(theta). Wheel power is approximately P_wheel = F_trac v, and battery power is higher after allowing for transmission and electrical losses.

> **Malayalam support:** Vehicle move □□□□□□□□ rolling resistance, aerodynamic drag, grade resistance, acceleration resistance □□□□□□ □□□□□□□□□□. Speed □□□□□□□□□ aerodynamic drag v^2 □□□□□□□□□□□ □□□□□□. Hill □□□□□□□□□ grade force □□□□□□□□□□□.

Study points

- Define each term in the tractive-force relation and give SI units.
- Explain why aerodynamic drag becomes significant at higher speed.
- Include transmission and electrical efficiency when moving from wheel power to battery power.

> **Engineering / workplace application:** The same force balance supports route simulation, motor sizing, energy prediction, and regenerative-braking estimation.

> **Illustrative example:** For a 1000 kg vehicle on a 5 degree slope, the grade component is approximately m g sin(5°), about 855 N before other resistances are added.

> **Common mistake:** Mixing vehicle mass in kilograms with force in newtons without multiplying by g gives an incorrect resistance.

Handbook treatment

M3.02 — Vehicle dynamics and performance (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.02 — Vehicle dynamics and performance (4 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Vehicle dynamics and performance (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Vehicle dynamics and performance (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on M3.02 — Vehicle dynamics and performance (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 — Vehicle dynamics and performance (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.02 — Vehicle dynamics and performance (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.02 — Vehicle dynamics and performance (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 — Vehicle dynamics and performance (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Vehicle dynamics and performance (4 hours)?
- How would you distinguish M3.02 — Vehicle dynamics and performance (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Vehicle dynamics and performance (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Vehicle dynamics and performance (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.02 — Vehicle dynamics and performance (4 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.03 — Classification and requirements of EV motors (4 hours)

Concept and explanation

EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, induction, permanent-magnet synchronous, and switched-reluctance families. Motor selection depends on torque-speed profile, inverter compatibility, cooling, fault tolerance, rare-earth dependence, and service requirements.

Malayalam support: EV motor □□ starting torque, wide speed range, high efficiency, regenerative operation, compact size, reliability □□□□□□ □□□□. Motor family □□□□□□□□□□□□□□□□□□□ torque-speed curve, inverter, cooling, cost, materials □□□□□□ □□□□□□□□□□□.

Study points

- List performance requirements of a traction motor.
- Classify motors by commutation or magnetic-field principle.
- Relate torque-speed characteristics to vehicle launch and cruising.

Engineering / workplace application: Traction motors are selected for the complete drive cycle rather than a single laboratory rating point.

Illustrative example: A motor can provide high low-speed torque for launch and then enter a constant-power region using field weakening or suitable control.

Common mistake: A high peak torque rating alone is not enough if the motor cannot dissipate heat during sustained hill climbing.

Handbook treatment

M3.03 — Classification and requirements of EV motors (4 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M3.03 — Classification and requirements of EV motors (4 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Classification and requirements of EV motors (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Classification and requirements of EV motors (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on M3.03 — Classification and requirements of EV motors (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 — Classification and requirements of EV motors (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M3.03 — Classification and requirements of EV motors (4 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M3.03 — Classification and requirements of EV motors (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 — Classification and requirements of EV motors (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Classification and requirements of EV motors (4 hours)?
- How would you distinguish M3.03 — Classification and requirements of EV motors (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Classification and requirements of EV motors (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Classification and requirements of EV motors (4 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M3.03 — Classification and requirements of EV motors (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.04 — Comparison of motors used in EVs (3 hours)

Concept and explanation

Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent magnets and can tolerate high speed, but their control and efficiency characteristics differ. Permanent-magnet motors offer high power density but depend on magnets and careful thermal control. Switched-reluctance motors are robust and magnet-free but can have torque ripple and acoustic noise.

Malayalam support: Brushed DC motor maintenance □□□□□□ □□□; BLDC electronic commutation □□□□□□□□□□□□□□. Induction motor magnets □□□□□□□ □□□□□□□□□□□□□□. Permanent-magnet motor power density □□□□□□□□; SR motor robust □□□□□□□□□□ torque ripple □□□□□□□□.

Study points

- Compare commutation, maintenance, power density, control, cost, and typical limitations.
- Use a table in examination answers when comparing motor types.
- State that the best motor depends on the vehicle duty cycle and system constraints.

Engineering / workplace application: Motor comparison supports the selection of traction machines for two-wheelers, passenger vehicles, buses, and industrial drives.

Illustrative example: For a compact vehicle requiring high power density, a permanent-magnet motor may be attractive; for a cost-sensitive design, an induction or switched-reluctance alternative may be evaluated.

Common mistake: Saying that one motor type is universally best ignores application, duty cycle, cost, and control constraints.

Handbook treatment

M3.04 — Comparison of motors used in EVs (3 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M3.04 — Comparison of motors used in EVs (3 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Comparison of motors used in EVs (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Comparison of motors used in EVs (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design Requirements, Dynamics and Motors.
- Write an examination answer on M3.04 — Comparison of motors used in EVs (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 — Comparison of motors used in EVs (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M3.04 — Comparison of motors used in EVs (3 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M3.04 — Comparison of motors used in EVs (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 — Comparison of motors used in EVs (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Comparison of motors used in EVs (3 hours)?
- How would you distinguish M3.04 — Comparison of motors used in EVs (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Comparison of motors used in EVs (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Comparison of motors used in EVs (3 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M3.04 — Comparison of motors used in EVs (3 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□.

Module summary: An EV design is a chain from performance requirement to force balance, power/energy requirement, transmission, and motor selection.

OFFICIAL MODULE 4

EV Braking, Sensors, Cooling, Service and Safety

This module focuses on the support systems that make an EV safe and usable: brake control, thermal management, sensors, maintenance, and electronic stability.

Hours: 15
CO4 · Bloom level: Understanding

Handbook study routine: Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 4

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

s:
Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking
system
Cooling Systems - EV batteries and motors
Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor
Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion
engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability
system.

Point-by-point study guide

Official syllabus point 1: Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking

Exact source point

> **Syllabus text:** Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking

How to understand and study it

This point is an official syllabus requirement: “Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking using the official terminology retained above.
- Organise the important elements of Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking into a logical sequence, classification, structure, or calculation.
- Connect Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking?
- How would you distinguish Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking from a closely related idea?
- Where would an engineer or technician encounter Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking, and what must be checked before using it?
- What common mistake would make an answer or operation involving Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Brake system for EV’s - Air Conditioning Systems, Adaptive cruise control, ABS braking □□□□□ topic □□□□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□□ □□□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□□. Exam answer □□□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□□.

Official syllabus point 2: Cooling Systems - EV batteries and motors

Exact source point

Syllabus text: Cooling Systems - EV batteries and motors

How to understand and study it

This point is an official syllabus requirement: “Cooling Systems - EV batteries and motors”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Cooling Systems - EV batteries, motors □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□ revision □□□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Cooling Systems - EV batteries and motors is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Cooling Systems - EV batteries and motors using the official terminology retained above.
- Organise the important elements of Cooling Systems - EV batteries and motors into a logical sequence, classification, structure, or calculation.
- Connect Cooling Systems - EV batteries and motors with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on Cooling Systems - EV batteries and motors with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Cooling Systems - EV batteries and motors. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Cooling Systems - EV batteries and motors is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Cooling Systems - EV batteries and motors, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Cooling Systems - EV batteries and motors, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Cooling Systems - EV batteries and motors?
- How would you distinguish Cooling Systems - EV batteries and motors from a closely related idea?
- Where would an engineer or technician encounter Cooling Systems - EV batteries and motors, and what must be checked before using it?
- What common mistake would make an answer or operation involving Cooling Systems - EV batteries and motors incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Cooling Systems - EV batteries and motors [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor

Exact source point

Syllabus text: Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor

How to understand and study it

This point is an official syllabus requirement: “Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Sensors used in EV’s- Temperature sensor, current sensors, coolant level sensor □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor using the official terminology retained above.
- Organise the important elements of Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor into a logical sequence, classification, structure, or calculation.
- Connect Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor?
- How would you distinguish Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor from a closely related idea?
- Where would an engineer or technician encounter Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor, and what must be checked before using it?
- What common mistake would make an answer or operation involving Sensors used in EV’s- Temperature sensor , MEMS, current sensors , coolant level sensor incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Sensors used in EV's- Temperature sensor , MEMS, current sensors , coolant level sensor [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion

Exact source point

Syllabus text: Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion

How to understand and study it

This point is an official syllabus requirement: “Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Maintenance, safety of EV, Electric vehicle maintenance as compared to combustion □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

Learning objectives

- Define and scope Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion using the official terminology retained above.
- Organise the important elements of Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion into a logical sequence, classification, structure, or calculation.
- Connect Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion. It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

Engineering application lens

In engineering practice, Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

Worked answer method

Given: the work activity, equipment, hazard, or incident condition.
Required: identify risk and propose controls.
Method: describe hazard → consequence → control → verification → emergency action.
Answer: give specific controls, responsible action, and one condition that must be checked before work begins.

Exam answer blueprint

For a short-answer question on Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion?
- How would you distinguish Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion from a closely related idea?
- Where would an engineer or technician encounter Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion, and what must be checked before using it?
- What common mistake would make an answer or operation involving Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion incorrect?

Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

Malayalam support: Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 5: engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability

Exact source point

Syllabus text: engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability

How to understand and study it

This point is an official syllabus requirement: “engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ engine, maintenance of electric motor, braking system ,battery ,sensors, electronic stability □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability using the official terminology retained above.
- Organise the important elements of engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability into a logical sequence, classification, structure, or calculation.
- Connect engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability?
- How would you distinguish engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability from a closely related idea?
- Where would an engineer or technician encounter engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability, and what must be checked before using it?
- What common mistake would make an answer or operation involving engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

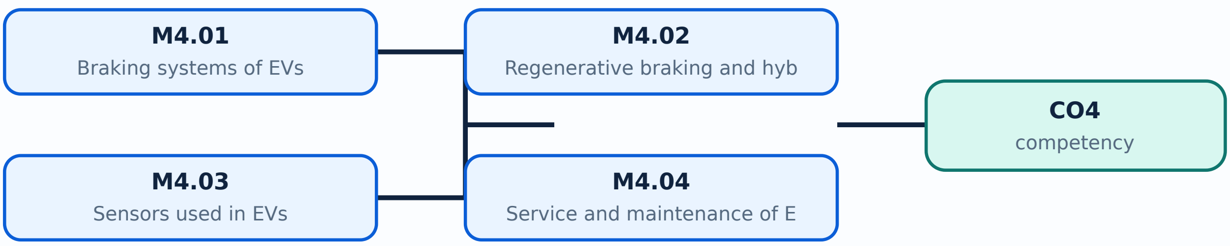
Malayalam support: engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Learning goals

- Describe braking, air-conditioning, adaptive cruise control, and ABS-related requirements.
- Explain regenerative and hybrid braking together with battery and motor cooling.
- Identify sensors and outline safe maintenance of EV subsystems.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Braking systems of EVs (3 hours)

Concept and explanation

An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-conditioning loads affect electrical energy consumption, while adaptive cruise control uses sensors and control logic to maintain speed or distance and can request propulsion or braking torque.

Malayalam support: EV brake system □ brake input, controller, actuator, friction brake, regenerative path, wheel-speed sensor □□□□□□ □□□□□□□□. ABS wheel lock □□□□□□□□□□ pressure modulate □□□□□□□. Air-conditioning energy consumption-□□ □□□□□□□□□.

Study points

- State the purpose of ABS and adaptive cruise control.
- Identify brake-system elements that must remain available under electrical faults.
- Relate auxiliary loads such as air conditioning to EV range.

Engineering / workplace application: Brake-by-wire and stability systems coordinate vehicle safety with energy recovery without compromising stopping performance.

Illustrative example: If a wheel-speed sensor detects incipient lock, ABS reduces and reapplies pressure while the vehicle controller manages any regenerative contribution.

Common mistake: Adaptive cruise control is not a substitute for driver attention or an independent guarantee of safe stopping.

Handbook treatment

M4.01 — Braking systems of EVs (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.01 — Braking systems of EVs (3 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — Braking systems of EVs (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — Braking systems of EVs (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on M4.01 — Braking systems of EVs (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 — Braking systems of EVs (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.01 — Braking systems of EVs (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.01 — Braking systems of EVs (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 — Braking systems of EVs (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — Braking systems of EVs (3 hours)?
- How would you distinguish M4.01 — Braking systems of EVs (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — Braking systems of EVs (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — Braking systems of EVs (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.01 — Braking systems of EVs (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.02 — Regenerative braking and hybrid braking (4 hours)

Concept and explanation

Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control system considers battery state of charge, motor speed, temperature, tyre adhesion, ABS demand, and brake-pedal feel. Regeneration is reduced when the battery cannot accept charge or when strong emergency braking is required.

Malayalam support: Motor generator ▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ kinetic energy electrical energy ▯▯▯▯▯ battery▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Battery limit, tyre grip, ABS demand ▯▯▯▯▯▯ ▯▯▯▯▯ friction brake ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ blended braking ▯▯▯▯▯▯▯▯▯▯.

Study points

- Explain the energy conversion during regeneration.
- List conditions that reduce regenerative braking.
- Describe how brake-pedal feel and stopping safety are preserved.

Engineering / workplace application: Regeneration is particularly useful in stop-start urban driving, downhill operation, and fleet vehicles.

Illustrative example: When the battery is near full charge, the controller reduces generator torque and relies more on friction braking while maintaining the requested deceleration.

Common mistake: Regenerative braking should not be treated as free energy; conversion losses and battery limits remain.

Handbook treatment

M4.02 — Regenerative braking and hybrid braking (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.02 — Regenerative braking and hybrid braking (4 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Regenerative braking and hybrid braking (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Regenerative braking and hybrid braking (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on M4.02 — Regenerative braking and hybrid braking (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 — Regenerative braking and hybrid braking (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.02 — Regenerative braking and hybrid braking (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.02 — Regenerative braking and hybrid braking (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 — Regenerative braking and hybrid braking (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Regenerative braking and hybrid braking (4 hours)?
- How would you distinguish M4.02 — Regenerative braking and hybrid braking (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Regenerative braking and hybrid braking (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Regenerative braking and hybrid braking (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.02 — Regenerative braking and hybrid braking (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.03 — Sensors used in EVs (4 hours)

Concept and explanation

Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protection, and charger monitoring. A coolant-level sensor detects insufficient coolant that could cause thermal damage. Sensor signals are filtered, plausibility-checked, and used by the vehicle controller for warnings, derating, or shutdown.

Malayalam support: Temperature sensor, MEMS, current sensor, coolant-level sensor □□□□□□ battery, motor, inverter, cooling system □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□. Sensor signal fault □□□□□□□□ controller warning □□□□□□□□□□ derating □□□□□□□.

Study points

- Match each sensor with its measured quantity and system use.
- Explain why redundant or plausibility checks are important for safety signals.
- Distinguish measurement, signal conditioning, and controller action.

Engineering / workplace application: Sensors are used in battery-management systems, motor drives, thermal control, driver assistance, and electronic stability systems.

Illustrative example: If coolant temperature rises above a calibrated limit, the controller can reduce power, command cooling, store a diagnostic code, and alert the driver.

Common mistake: A sensor reading should not be trusted blindly; open-circuit, short-circuit, out-of-range, and implausible values require diagnosis.

Handbook treatment

M4.03 — Sensors used in EVs (4 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope M4.03 — Sensors used in EVs (4 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — Sensors used in EVs (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — Sensors used in EVs (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on M4.03 — Sensors used in EVs (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 — Sensors used in EVs (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of M4.03 — Sensors used in EVs (4 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.
Required: a suitable measurement chain or an explanation of how the instrument operates.
Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on M4.03 — Sensors used in EVs (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 — Sensors used in EVs (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — Sensors used in EVs (4 hours)?
- How would you distinguish M4.03 — Sensors used in EVs (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — Sensors used in EVs (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — Sensors used in EVs (4 hours) incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: M4.03 — Sensors used in EVs (4 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

M4.04 — Service and maintenance of EV components (4 hours)

Concept and explanation

EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, battery and cooling-system checks, motor and inverter inspection, brake and tyre service, sensor checks, electronic-stability diagnostics, and safe handling of charging equipment. Only trained personnel should work on high-voltage circuits using approved isolation, PPE, insulated tools, and lockout procedures.

> **Malayalam support:** EV maintenance-▯ engine oil/exhaust service ▯▯▯▯▯▯▯▯▯▯ high-voltage safety ▯▯▯▯▯▯▯▯▯▯▯. Battery, cooling, motor, inverter, brakes, tyres, sensors, charging equipment ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯. Trained person ▯▯▯▯▯▯ HV circuit service ▯▯▯▯▯▯▯.

Study points

- Compare EV maintenance with combustion-engine maintenance.
- List high-voltage safety steps before service.
- Include battery, motor, brakes, sensors, cooling, and electronic stability in a maintenance plan.

> **Engineering / workplace application:** Service procedures are applied in workshops, fleet depots, charging stations, and accident-recovery situations.

> **Illustrative example:** A safe service sequence is identify the vehicle, switch off, isolate and lock out the high-voltage source, wait the specified discharge time, verify absence of voltage, then begin work.

> **Common mistake:** Disconnecting a visible connector does not by itself prove that a high-voltage system is safe; isolation must be verified according to procedure.

Handbook treatment

M4.04 — Service and maintenance of EV components (4 hours) must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

Learning objectives

- Define and scope M4.04 — Service and maintenance of EV components (4 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — Service and maintenance of EV components (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — Service and maintenance of EV components (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Braking, Sensors, Cooling, Service and Safety.
- Write an examination answer on M4.04 — Service and maintenance of EV components (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 — Service and maintenance of EV components (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

Engineering application lens

In engineering practice, M4.04 — Service and maintenance of EV components (4 hours) is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

Worked answer method

Given: the work activity, equipment, hazard, or incident condition.
Required: identify risk and propose controls.
Method: describe hazard → consequence → control → verification → emergency action.
Answer: give specific controls, responsible action, and one condition that must be checked before work begins.

Exam answer blueprint

For a short-answer question on M4.04 — Service and maintenance of EV components (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 — Service and maintenance of EV components (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — Service and maintenance of EV components (4 hours)?
- How would you distinguish M4.04 — Service and maintenance of EV components (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — Service and maintenance of EV components (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — Service and maintenance of EV components (4 hours) incorrect?

Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

Malayalam support: M4.04 — Service and maintenance of EV components (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: Reliable EV operation depends on coordinated brakes, sensors, cooling, maintenance, and safety procedures, not only on the traction motor.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

learning-map diagram.

Module 2: EV Control

rning-map diagram.

Module 3: EV Design

rning-map diagram.

Module 4: EV Braking,

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Source anomaly: The supplied course outline lists Series Test I and Series Test II as 1 hour within module entries, while the Course Outcomes table separately lists Series Test as 2 hours.
- Source anomaly: The module table gives Module 4 a total of 15 hours, while CO4 states 13 hours; the 1-hour Series Test II is also shown under Module 4. These official values are preserved without silent correction.
- No separate CIE/ESE mark distribution is stated in the supplied PDF.

Module-wise Questions and Answers

module-1 — Electric and Hybrid Vehicle Fundamentals

1. Explain Need for hybrid and electric vehicles. **[3 marks]**

Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and the possibility of using electricity from diverse sources. Historical development moved from early battery vehicles through modern power electronics, lithium-based batteries, charging infrastructure, and electronically controlled propulsion. Social importance includes quieter transport and reduced local pollution; environmental benefit depends on the electricity-generation mix, battery manufacture, and responsible end-of-life management.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Components and working principles of electric and hybrid vehicles. **[3 marks]**

A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loads, and vehicle control unit. An HEV adds an engine, fuel system, generator or motor-generator, and a supervisory energy-management controller. During propulsion, stored electrical energy passes through the inverter to the motor; during regenerative deceleration, the power flow reverses and the motor operates as a generator.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Configurations of Electric Vehicles. **[3 marks]**

The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An FCEV uses a fuel cell to generate electricity, usually with a battery or buffer storage. An HEV combines an engine and an electrical path, while a PHEV adds a larger rechargeable battery and external charging capability. EVs may also be classified by whether torque is transmitted through a differential or through separate in-wheel or near-wheel drives.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Types of hybrid electric vehicle. **[3 marks]**

In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split device. A complex hybrid can include multiple motors, clutches, or flexible power paths. An Extended Range Electric Vehicle (EREV) primarily uses the electric drive for wheel propulsion while an engine-generator extends the available range when required.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — EV Control Systems, Drive Trains and Braking

5. Explain Control systems for EVs. **[3 marks]**

An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-electronics rating, energy-storage selection, communications network, and supporting subsystems such as cooling, steering, lighting, and braking. The controller must enforce voltage, current, temperature, and isolation limits while meeting the requested vehicle torque.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Split power devices for HEV. **[3 marks]**

A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator and motor torque so the engine can operate efficiently while the vehicle meets its speed and acceleration demand.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Drive trains in EV and HEV. **[3 marks]**

An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery, fuel-cell, hybrid, or multiple-source power. Single-motor drives are simpler; multi-motor drives can distribute torque between axles or wheels. In-wheel drives reduce mechanical transmission parts but increase wheel mass and expose the motor to harsh conditions.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Braking systems in EV. [3 marks]

EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provide the remaining deceleration and hold the vehicle at rest. Hybrid braking coordinates pedal feel, motor torque, hydraulic pressure, ABS, and stability control. The braking system must remain effective when the battery is full, cold, overheated, or unable to accept regeneration.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — EV Design Requirements, Dynamics and Motors

9. Explain Design requirements for electric vehicles. [3 marks]

Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy storage can increase mass, while higher acceleration demands higher peak motor and inverter power. A design statement should define the duty cycle, road conditions, payload, ambient temperature, and allowable state-of-charge window before component ratings are selected.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Vehicle dynamics and performance. [3 marks]

The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is F_trac = F_roll + F_aero + F_grade + m a. Rolling resistance may be approximated by C_rr m g; aerodynamic drag by 0.5 rho C_d A v^2; grade force by m g sin(theta). Wheel power is approximately P_wheel = F_trac v, and battery power is higher after allowing for transmission and electrical losses.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Classification and requirements of EV motors. [3 marks]

EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, induction, permanent-magnet synchronous, and switched-reluctance families. Motor selection depends on torque-speed profile, inverter compatibility, cooling, fault tolerance, rare-earth dependence, and service requirements.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Comparison of motors used in EVs. **[3 marks]**

Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent magnets and can tolerate high speed, but their control and efficiency characteristics differ. Permanent-magnet motors offer high power density but depend on magnets and careful thermal control. Switched-reluctance motors are robust and magnet-free but can have torque ripple and acoustic noise.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — EV Braking, Sensors, Cooling, Service and Safety

13. Explain Braking systems of EVs. **[3 marks]**

An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-conditioning loads affect electrical energy consumption, while adaptive cruise control uses sensors and control logic to maintain speed or distance and can request propulsion or braking torque.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Regenerative braking and hybrid braking. **[3 marks]**

Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control system considers battery state of charge, motor speed, temperature, tyre adhesion, ABS demand, and brake-pedal feel. Regeneration is reduced when the battery cannot accept charge or when strong emergency braking is required.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Sensors used in EVs. **[3 marks]**

Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protection, and charger monitoring. A coolant-level sensor detects insufficient coolant that could cause thermal damage. Sensor signals are filtered, plausibility-checked, and used by the vehicle controller for warnings, derating, or shutdown.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Service and maintenance of EV components. **[3 marks]**

EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, battery and cooling-system checks, motor and inverter inspection, brake and tyre service, sensor checks, electronic-stability diagnostics, and safe handling of charging equipment. Only trained personnel should work on high-voltage circuits using approved isolation, PPE, insulated tools, and lockout procedures.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need for hybrid and electric vehicles* with one relevant application. (5 marks)
2. **2.** Define or explain *Components and working principles of electric and hybrid vehicles* with one relevant application. (5 marks)
3. **3.** Define or explain *Control systems for EVs* with one relevant application. (5 marks)
4. **4.** Define or explain *Split power devices for HEV* with one relevant application. (5 marks)
5. **5.** Define or explain *Design requirements for electric vehicles* with one relevant application. (5 marks)
6. **6.** Define or explain *Vehicle dynamics and performance* with one relevant application. (5 marks)
7. **7.** Define or explain *Braking systems of EVs* with one relevant application. (5 marks)
8. **8.** Define or explain *Regenerative braking and hybrid braking* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Electric and Hybrid Vehicle Fundamentals:** EV and HEV classification becomes clear when the student follows the energy source, conversion path, and wheel-torque path.
- **EV Control Systems, Drive Trains and Braking:** Control, drive-train, and brake decisions are coupled: the vehicle must satisfy torque and safety requirements while managing energy flow.
- **EV Design Requirements, Dynamics and Motors:** An EV design is a chain from performance requirement to force balance, power/energy requirement, transmission, and motor selection.
- **EV Braking, Sensors, Cooling, Service and Safety:** Reliable EV operation depends on coordinated brakes, sensors, cooling, maintenance, and safety procedures, not only on the traction motor.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Need for hybrid and electric vehicles	Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and t
Components and working principles of electric and hybrid vehicles	A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loa
Configurations of Electric Vehicles	The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An
Types of hybrid electric vehicle	In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split d
Control systems for EVs	An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-el
Split power devices for HEV	A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator a

Term	Short meaning
Drive trains in EV and HEV	An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery,
Braking systems in EV	EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provid
Design requirements for electric vehicles	Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy sto
Vehicle dynamics and performance	The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is $F_{trac} = F_{roll} + F_{aero} + F_{grade} + m a$. Rolling resistance may be approximated b
Classification and requirements of EV motors	EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, inducti
Comparison of motors used in EVs	Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent
Braking systems of EVs	An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-condi
Regenerative braking and hybrid braking	Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control s
Sensors used in EVs	Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protec
Service and maintenance of EV components	EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, b

Official References and Resources

References listed in the supplied syllabus

1. Jack Erjavec and Jeff Arias, Hybrid, Electric and Fuel-cell Vehicles, Cengage Learning India Pvt Ltd., New Delhi.
2. S. Onori, L. Serrao and G. Rizzoni, Hybrid Electric Vehicles: Energy Management Strategies, Springer, 2015.
3. T. Denton, Electric and Hybrid Vehicles, Routledge.
4. A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi.
5. C. Mi, M. A. Masrur and D. W. Gao, Hybrid Electric Vehicles: Principles and Applications with ... [the supplied PDF truncates the title here].

Official learning resources listed

- https://nptel.ac.in/courses/108/103/108103009/
- https://nptel.ac.in/courses/108/106/108106170
-
-
- **In-page Search**